

ML Scan Parameters

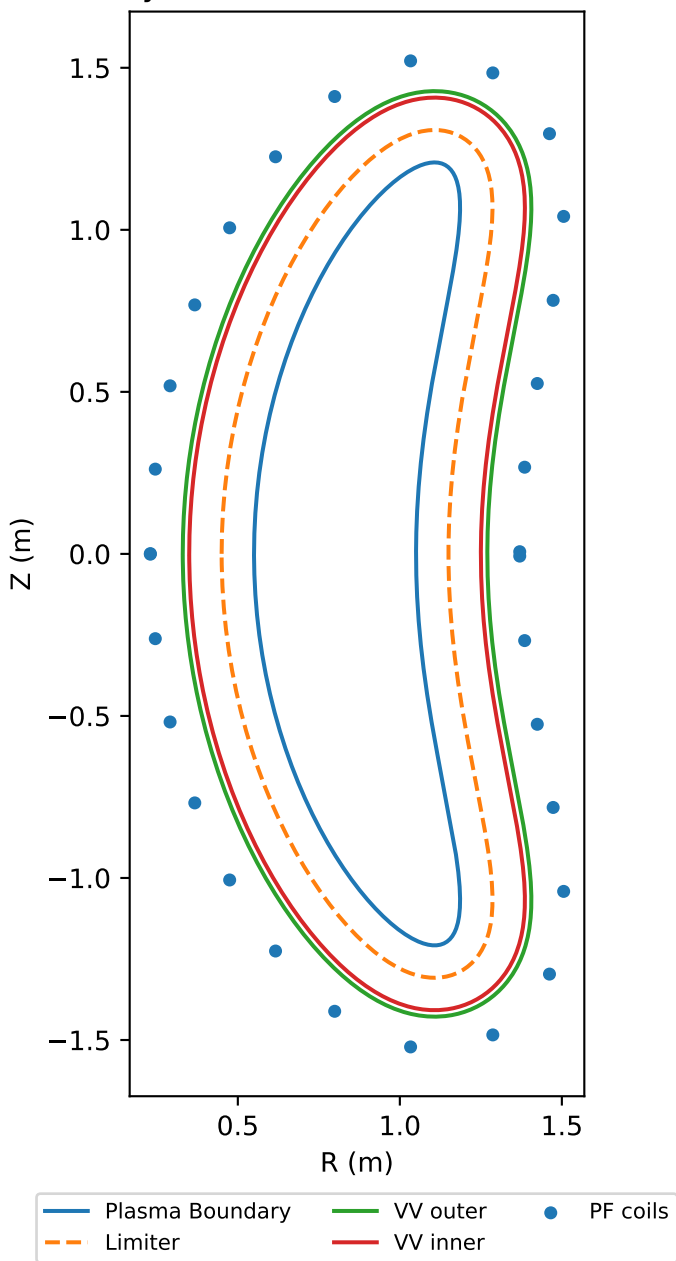
C-shape plasma $(r,z) = (A + B\sin t + C\cos 2t, F + G\cos t - H\sin 2t)$

A=0.9700 B=0.2500 C=0.1700
G=1.2000 H=0.0700 F=0.0
R0=0.8000 m kappa_target=4.800 delta=-0.070
B0=1.8 T Ip=1.00 MA

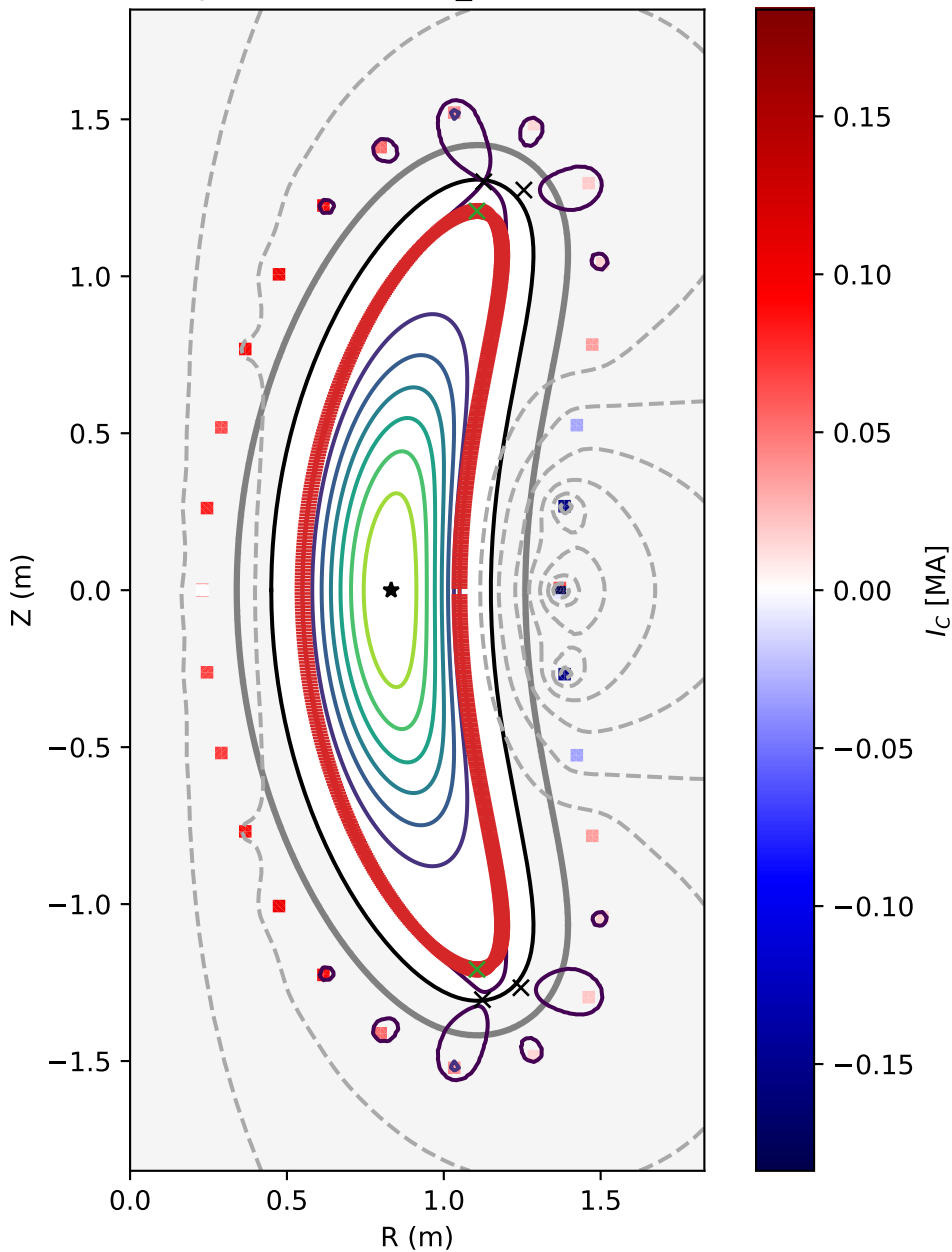
Plasma profiles:
pp_prof: $(1 - \psi^{0.599})^{0.823}$
ffp_prof: $(1 - \psi^{2.410})^{2.130}$

Scan params: B=2.5 C=17 G=12 H=7
alpha_pp=0.599 gamma_pp=0.823
alpha_ff=2.410 gamma_ff=2.130

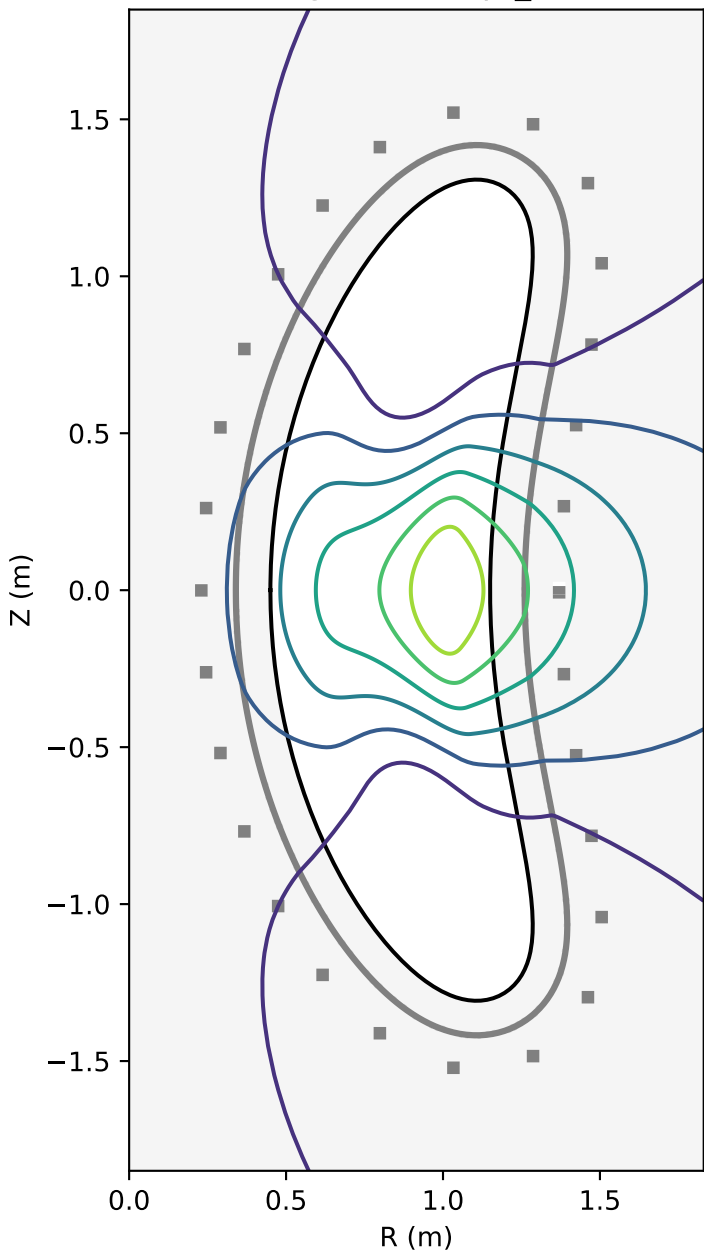
Geometry B=2.5, C=17, G=12, H=7, $\kappa=4.8$



Equilibrium flux $\kappa_{\text{actual}}=3.98$



Unstable eigenmode $\gamma\tau_w = -1.000$



Vertical Stability Analysis

Vertical Stability Results

=====

B=2.5 C=17 G=12 H=7
alpha_pp=0.599 gamma_pp=0.823
alpha_ff=2.410 gamma_ff=2.130

kappa_target = 4.800
kappa_actual = 3.983
beta_N = 0.470 %

growth rate (gamma) = -5.0268E+01 s-1
wall time (tau_w) = 1.9894E-02 s
gamma * tau_w = -1.0000
max coil current = 0.184 MA
Category: passive stable

axis_offset_pct = 4.02%

Equilibrium Statistics

Equilibrium Statistics:	
Topology	= Diverted
Toroidal Current [A]	= 9.9999E+05
Current Centroid [m]	= 0.848 0.000
Magnetic Axis [m]	= 0.832 0.000
Elongation	= 3.983 (U: 4.223, L: 3.743)
Triangularity	= -0.783 (U: -0.869, L: -0.697)
Plasma Volume [m^3]	= 5.885
q_0, q_95	= 2.403 9.838
Peak Pressure [Pa]	= 5.1714E+04
Stored Energy [J]	= 8.9764E+04
<Beta_pol> [%]	= 41.9359
<Beta_tor> [%]	= 0.9108
<Beta_n> [%]	= 0.4700
Diamagnetic flux [Wb]	= 8.3079E-03
Toroidal flux [Wb]	= 1.7654E+00
l_i	= 0.8659